

H.S. 2nd Year — Computer Science and Applications (2025)

Solved Question Paper — Code: 36 / CSCA

Full Marks: 70 | Pass Marks: 21 | Time: Three hours

Q. No. 1 — Answer any eight ($1 \times 8 = 8$) [All parts solved]

(a) What is OOP ?

Ans: OOP stands for Object-Oriented Programming — a programming paradigm based on the concept of 'objects', which contain data (attributes) and code (methods), and which uses features like class, encapsulation, inheritance, and polymorphism to design programs.

(b) What is an instance in OOP ?

Ans: An instance is another term for an object — a specific realisation of a class, created in memory with actual values for its data members. Creating an object from a class is called 'instantiating' the class.

(c) What is Data Encapsulation ?

Ans: Data Encapsulation is the OOP mechanism of binding data (data members) and the functions that operate on that data (member functions) together into a single unit (class), while restricting direct access to some of the object's components (data hiding), thereby protecting data from unauthorised access.

(d) What is Overloading ?

Ans: Overloading is a feature of C++ that allows a function name or an operator to be used for more than one purpose/task. It includes function overloading (same function name, different parameters) and operator overloading (redefining the meaning of an operator for user-defined types). It is a form of compile-time polymorphism.

(e) What is Destructor in OOP ?

Ans: A destructor is a special member function with the same name as its class, preceded by a tilde (~), with no return type or arguments, which is automatically invoked when an object is destroyed, to release the resources/memory used by that object.

(f) What is class member ?

Ans: A class member refers to any data member (variable) or member function declared inside a class, which together define the properties and behaviour of the objects created from that class.

(g) What is Relational Algebra ?

Ans: Relational Algebra is a procedural query language used in RDBMS, which takes one or more relations (tables) as input and produces a new relation as output using operations like SELECT, PROJECT, JOIN and UNION, to retrieve and manipulate data.

(h) What is an Array ?

Ans: An array is a collection of elements of the same data type, stored in contiguous (consecutive) memory locations, and accessed using a common name along with an index/subscript, e.g., `int a[5];`

(i) What is Memory Allocation ?

Ans: Memory allocation is the process of reserving a portion of computer memory (RAM) for storing data or program instructions during execution. It can be static (fixed size, allocated at compile time) or dynamic (allocated during run time using operators like 'new').

(j) What does FIFO mean to you ?

Ans: FIFO stands for 'First In First Out'. It is the principle followed by a Queue, in which the element inserted first is the first one to be removed/deleted.

(k) What is preprocessor directive ?

Ans: A preprocessor directive is an instruction to the compiler's preprocessor, given before the main program compiles, usually starting with the '#' symbol, e.g., #include, #define. These are processed before actual compilation begins, e.g., #include <iostream> includes the header file needed for input/output.

(l) What is Relational Database ?

Ans: A relational database is a type of database that stores and organises data in the form of related tables (rows and columns), where data in different tables can be linked using common fields (keys), following the relational model proposed by E. F. Codd.

(m) What is Topology ?

Ans: Topology (network topology) refers to the physical or logical arrangement/layout of computers, cables, and other devices that make up a network, e.g., bus, star, ring, mesh topology.

Q. No. 2 — Answer any ten ($2 \times 10 = 20$) [All parts solved]

(a) Define Namespaces.

Ans: A namespace in C++ is a declarative region that provides a scope to identifiers (names of variables, functions, classes, etc.) to avoid naming conflicts, especially in large programs. The standard namespace used in C++ is 'std' (as in 'using namespace std;'), which contains standard library elements like cout and cin.

(b) Define Sorting.

Ans: Sorting is the process of arranging the elements of a list/array in a particular order — either ascending or descending — based on some criterion (e.g., numeric value or alphabetical order), e.g., using techniques like bubble sort, insertion sort, or selection sort.

(c) Define POSTFIX notation.

Ans: Postfix notation (also called Reverse Polish Notation) is a way of writing an expression in which the operator is placed after its operands, e.g., the infix expression $A + B$ is written as $AB+$ in postfix. It removes the need for brackets and operator precedence rules while evaluating expressions using a stack.

(d) State De-Morgan's theorems.

Ans: De Morgan's Theorems in Boolean algebra state: (1) $(A + B)' = A'.B'$ — the complement of a sum is the product of the complements. (2) $(A.B)' = A' + B'$ — the complement of a product is the sum of the complements.

(e) What is the use of Optical Fibre ?

Ans: Optical fibre is a transmission medium that carries data as pulses of light through thin strands of glass or plastic. It is used for high-speed, long-distance data transmission with very high bandwidth and minimal signal loss/interference, e.g., in internet backbones, telecommunication networks, and cable TV.

(f) Describe Satellite Data Communication of Networking.

Ans: Satellite data communication is a method of wireless communication in which data is transmitted between an earth station and a communication satellite orbiting the earth, which then relays the signal to another earth station. It is used to provide network connectivity over very long distances and in remote areas where laying physical cables is difficult, e.g., for television broadcasting, GPS, and long-distance telephone/internet links.

(g) What can a template in C++ do ?

Ans: A template in C++ allows a programmer to write a single generic function or class that can work with any data type, without rewriting the code separately for each type. It enables generic programming, e.g., a template function can add two integers, two floats, or two doubles using the same code.

(h) Define a Queue.

Ans: A queue is a linear data structure that follows the FIFO (First In First Out) principle, where insertion (enqueue) takes place at the rear end and deletion (dequeue) takes place at the front end.

(i) What is IOSTREAM ?

Ans: IOSTREAM is a standard header file in C++ (<iostream>) that provides facilities for standard input and output operations, defining objects such as cin (for input) and cout (for output), along with related classes like istream and ostream.

(j) What is Wireless Communication ?

Ans: Wireless communication is the transmission of data/information between two or more devices without the use of physical wires or cables, using electromagnetic waves (such as radio waves, microwaves, or infrared) as the transmission medium, e.g., Wi-Fi, Bluetooth, mobile networks.

(k) Describe Bandwidth.

Ans: Bandwidth is the maximum rate of data transfer across a network/communication channel within a given time, usually measured in bits per second (bps). Higher bandwidth allows more data to be transmitted in the same amount of time, resulting in faster network performance.

(l) What is K-map simplification ?

Ans: K-map (Karnaugh Map) simplification is a graphical technique used to simplify Boolean/logical expressions without applying lengthy Boolean algebra laws, by grouping adjacent cells containing 1s (or 0s) in a specially arranged grid, to obtain the minimal SOP or POS expression.

(m) What is the use of XML ?

Ans: XML (eXtensible Markup Language) is used to store, structure, and transport data between different systems/applications in a platform-independent format, using user-defined custom tags. It focuses on describing what the data is, rather than how it should be displayed.

(n) What is Boolean algebra ?

Ans: Boolean algebra is a branch of mathematics that deals with variables that can take only two values — TRUE (1) and FALSE (0) — and operations like AND, OR, and NOT performed on them. It is widely used in designing and simplifying digital logic circuits.

(o) What is TCP/IP ?

Ans: TCP/IP (Transmission Control Protocol/Internet Protocol) is the standard suite of communication protocols used to connect devices over the Internet and other networks — TCP ensures reliable, ordered delivery of data, while IP handles addressing and routing of data packets.

Q. No. 3 — Answer any nine ($3 \times 9 = 27$) [All parts solved]

(a) Explain Constructor Overloading.

Ans: Constructor overloading is a feature of C++ in which a class has more than one constructor, each with a different number or type of parameters. The appropriate constructor is automatically called by the compiler depending on the arguments passed while creating an object. Example: `class Box { public: Box() {...} Box(int a) {...} Box(int a, int b) {...} };` — here three constructors of the same class 'Box' are overloaded, allowing objects to be created with no arguments, one argument, or two arguments as needed.

(b) Write the differences between Top-Down and Bottom-Up approaches.

Ans: Top-Down approach: (1) It starts with the main/overall problem and breaks it down into smaller sub-problems/modules step by step. (2) It follows a design method typical of procedural/structured programming. Bottom-Up approach: (1) It starts by designing and building the smallest, most basic components/modules first, and then combines them to build larger and more complex systems. (2) It is the typical approach followed in Object-Oriented Programming, where individual objects/classes are built first and then integrated.

(c) Explain the POP and PUSH operations in Stack.

Ans: PUSH operation: It is used to insert/add a new element onto the top of the stack. Before pushing, it is checked whether the stack is full (stack overflow condition); if not full, the top pointer is incremented and the new element is placed at that position. POP operation: It is used to remove/delete the element present at the top of the stack. Before popping, it is checked whether the stack is empty (stack underflow condition); if not empty, the top element is removed and the top pointer is decremented. Both PUSH and POP operations follow the LIFO (Last In First Out) principle.

(d) Describe the Queue and its operations.

Ans: A queue is a linear data structure that follows the FIFO (First In First Out) principle, where the element inserted first is removed first. Its main operations are: (1) Enqueue – inserting a new element at the rear end of the queue. (2) Dequeue – removing an element from the front end of the queue. (3) Peek/Front – viewing the front element without removing it. (4) isEmpty() – checking whether the queue has no elements. (5) isFull() – checking whether the queue (array-based) has reached its maximum capacity.

(e) Explain the Insertion Sorting.

Ans: Insertion sort is a simple sorting technique that builds the final sorted array one element at a time. It works by taking each element from the unsorted part of the array and inserting it into its correct position among the already-sorted part (to its left), shifting the larger elements one position to the right to make space. It is efficient for small or nearly-sorted data sets, with a worst-case time complexity of $O(n^2)$.

(f) Define the column constraints in DBMS.

Ans: Column constraints in DBMS are rules applied to a column of a table to limit the type of data that can be stored in it, ensuring accuracy and reliability of the data. Common constraints include: NOT NULL (column cannot have a null value), UNIQUE (all values in the column must be different), PRIMARY KEY (uniquely identifies each record), FOREIGN KEY (links to a primary key in another table), CHECK (restricts values based on a condition), and DEFAULT (sets a default value).

(g) Explain the Client side scripting.

Ans: Client-side scripting refers to scripts (programs) that are executed on the user's own web browser (client machine) rather than on the web server. These scripts are used to make web pages interactive/dynamic without sending requests to the server for every action, e.g., form validation, animations, and interactive UI elements. Common client-side scripting languages: JavaScript, VBScript. The source code is visible to the end user (viewable via 'View Page Source').

(h) Write about the logic gates.

Ans: Logic gates are the basic building blocks of digital circuits, which perform logical (Boolean) operations on one or more binary inputs to produce a single binary output. The basic logic gates are: AND gate (output is 1 only when all inputs are 1), OR gate (output is 1 when at least one input is 1), NOT gate (inverts the input), and the derived gates NAND, NOR, XOR, and XNOR. They are used to design and build digital devices such as computers and calculators.

(i) Explain HTML.

Ans: HTML (HyperText Markup Language) is the standard markup language used to create and design the structure/layout of web pages. It uses a set of predefined 'tags' (e.g., <html>, <head>, <body>, <p>,) to define different elements of a web page, which are then interpreted and displayed by a web browser. HTML pages can also be combined with CSS (for styling) and JavaScript (for interactivity).

(j) Define DDL and DML.

Ans: DDL (Data Definition Language): It is the subset of SQL commands used to define and modify the structure of database objects such as tables, e.g., CREATE, ALTER, DROP, TRUNCATE. DML (Data Manipulation Language): It is the subset of SQL commands used to manipulate/manage the actual data stored within database tables, e.g., INSERT, UPDATE, DELETE, SELECT.

(k) Mention the uses of Co-axial cable.

Ans: A co-axial cable is a wired transmission medium consisting of a central copper conductor surrounded by an insulating layer, a metallic shield, and an outer insulating jacket. It is used: (1) In cable TV networks to transmit television signals, (2) In broadband internet connections (e.g., cable internet), (3) In LANs (older Ethernet networks), and (4) For connecting antennas to television/radio receivers, because it offers better shielding against electromagnetic interference and higher bandwidth than twisted pair cable.

(l) Differentiate one and two dimensional arrays.

Ans: One-dimensional array: (1) It stores a list of elements in a single row (single index), e.g., `int a[5]`; (2) It is used to represent simple linear collections of data, like a list of marks. Two-dimensional array: (1) It stores elements in the form of rows and columns (two indices), e.g., `int a[3][4]`; (2) It is used to represent tabular/matrix data, like a grid or a table.

(m) Write about switching techniques.

Ans: Switching techniques are methods used in computer networks to transfer data from a source to a destination across intermediate nodes. The main types are: (1) Circuit Switching – a dedicated physical communication path/circuit is established between sender and receiver for the entire duration of communication (e.g., traditional telephone networks). (2) Message Switching – the entire message is sent as a single unit from node to node, being stored and then forwarded (store-and-forward) until it reaches the destination. (3) Packet Switching – the message is broken into smaller units called packets, which are sent independently (possibly via different routes) and reassembled at the destination; this is the technique used in the modern Internet.

Q. No. 4 — Answer any three ($5 \times 3 = 15$) [All parts solved]

(a) Explain different clauses used in SQL.

Ans: Common clauses used in SQL (mainly with the SELECT statement): (1) WHERE clause – used to filter records based on a specified condition, e.g., `SELECT * FROM Student WHERE Marks > 50`; (2) ORDER BY clause – used to sort the result set in ascending (ASC) or descending (DESC) order, e.g., `ORDER BY Marks DESC`; (3) GROUP BY clause – used to group rows that have the same values in specified columns, usually together with aggregate functions, e.g., `GROUP BY Department`; (4) HAVING clause – used to filter groups of records (created by GROUP BY) based on a condition, e.g., `HAVING COUNT(*) > 5`; (5) FROM clause – specifies the table(s) from which to retrieve data. These clauses together allow flexible retrieval, filtering, grouping and sorting of data stored in tables.

(b) Explain about DOMAIN names.

Ans: A domain name is a human-readable address used to identify a website or a server on the Internet (e.g., `www.example.com`), used in place of its numeric IP address so that it is easier for users to remember and access. It is translated into the corresponding IP address by the Domain Name System (DNS). A domain name typically consists of a top-level domain (TLD) such as `.com`, `.org`, `.edu`, `.gov`, or `.in`, and a second-level domain (the unique name registered by an organisation, e.g., 'example'). Domain names must be registered and are managed by domain registrars.

(c) Explain the Boolean operators.

Ans: Boolean operators are the basic operations performed on Boolean variables (which take values 0 or 1) in Boolean algebra. The three basic Boolean operators are: (1) AND (.) – gives output 1 only when both/all inputs are 1 (logical multiplication). (2) OR (+) – gives output 1 when at least one input is 1 (logical addition). (3) NOT (') – inverts/complements the input, i.e., changes 1 to 0 and 0 to 1 (logical negation). These three basic operators can be combined to form derived operators like NAND, NOR, XOR, and XNOR, and are used to design and simplify digital logic circuits.

(d) Write a program in C++ to sort the numbers entered by the user.

```
#include <iostream>
using namespace std;
int main()
{
```

```

int n, i, j, temp;
cout << "Enter number of elements: "; cin >> n;
int a[n];
cout << "Enter " << n << " numbers:\n";
for (i = 0; i < n; i++)
    cin >> a[i];
// Bubble sort
for (i = 0; i < n - 1; i++)
{
    for (j = 0; j < n - i - 1; j++)
    {
        if (a[j] > a[j + 1])
        {
            temp = a[j];
            a[j] = a[j + 1];
            a[j + 1] = temp;
        }
    }
}
cout << "\nSorted numbers (ascending order):\n";
for (i = 0; i < n; i++)
    cout << a[i] << " ";
return 0;
}

```

(e) Explain the difference between traversal and searching.

Ans: Traversal: (1) It is the process of visiting/accessing every element of a data structure (such as an array, linked list, or tree) exactly once, usually to process or display each element. (2) It always processes the entire data structure, e.g., in-order traversal of a tree visits every node. Searching: (1) It is the process of finding whether a particular element (key) is present in a data structure, and if so, locating its position. (2) It may not need to visit every element — it can stop as soon as the required element is found, e.g., in binary search.

(f) What are the differences between SMTP and POP3 protocols ?

Ans: SMTP (Simple Mail Transfer Protocol): (1) It is used to send (push) e-mail messages from a client to a mail server, or between mail servers. (2) It works on port 25 (default) and operates in a 'push' manner. POP3 (Post Office Protocol version 3): (1) It is used to retrieve (pull/download) e-mail messages from a mail server to the client's device. (2) It works on port 110 (default) and typically downloads and, by default, removes the mail from the server after retrieval. In short: SMTP is used for sending mail, while POP3 is used for receiving/downloading mail.